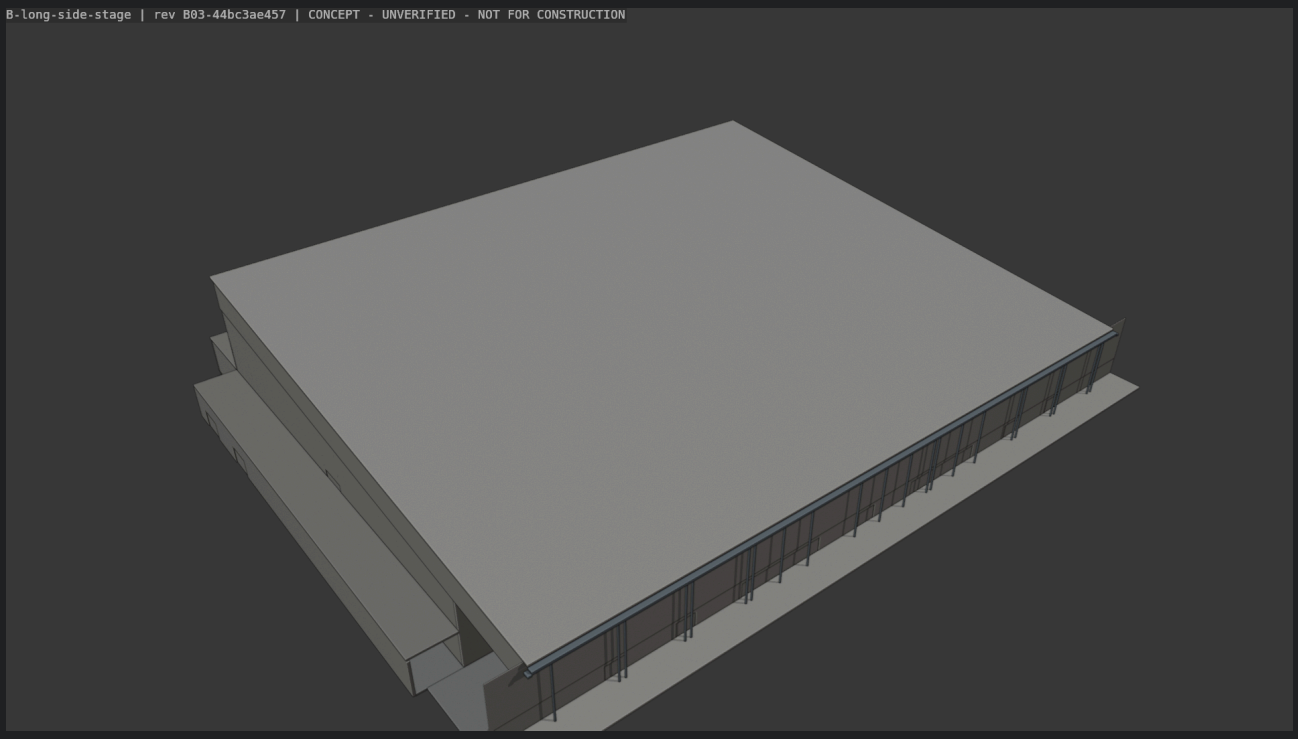
PRELIMINARY ARCHITECTURAL & STRUCTURAL ENGINEERING DESIGN PROPOSAL

Proposed 3,000-seat church auditorium Option B — long-side stage (wide, shallow fan)



DESIGN ID B-LONG-SIDE-STAGE · MODEL REVISION B03-44BC3AE457 · ZIMBABWE (SITE NOT IDENTIFIED) · 2026-11-02

CONCEPT — UNVERIFIED · NOT FOR CONSTRUCTION. Reviewer scores are QA metrics, not certification.

Document control & provenance

Design ID	B-long-side-stage
Model revision	B03-44bc3ae457
Manifest SHA-256	356464e19ccc094f284d8ae1a5c64159700c0fbf518a867ca8b60c1ce1169c83
Blender	5.2.3 LTS
Units	metres
Status	CONCEPT - UNVERIFIED - NOT FOR CONSTRUCTION
Build	.venv/Scripts/python tools/build.py --design B --iteration N
Measurement provenance	Every figure, table and drawing in this document is generated from this design's Blender manifest, its verified reports (geometry_checks, blend_verification, ifc_validation) and comparisons/compare.py run on the same manifest.

Source files hashed in the revision record

Source	SHA-256 (16)
config/params.json	f81053ac3ae5b801
src/layout.py	25bdca4a02ee1d70
src/reports.py	5af693842eefe319
src/drawings.py	0c8ec145e5044f22
tests/test_layout.py	e8b23aba493651eb
../../shared/bim/geometry.py	e24a0b2f94edac55
../../shared/bim/build_blender.py	95c6a9deb318822e
../../shared/bim/verify_blend.py	a8b887a647d43a18
../../shared/bim/export_ifc.py	63e72c166381e548
../../shared/bim/validate_ifc.py	afc360f18265ebdb
../../shared/drawings/sheets.py	e08e3a20e6083ee3
../../shared/pipeline.py	99b11afba30aadd2
../../shared/render_views.py	78894ec01a471503

Concept and scope

The platform stands on the middle of the long north wall. 3,000 seats sit in a wide, shallow fan of 32 concentric rows facing the platform focus; one mono-pitch roof covers foyer and hall.

Carried from the common brief / client preferences

- **seating**: Seat, row and aisle dimensions carried unchanged from Option A (common basis) so the comparison is like-for-like.
- **sightlines**: C-value targets, focal-point rule and riser limits carried from Option A (common basis).
- **accessibility**: Wheelchair-space size and 1 % concept target carried from Option A; locations are a NEW PROPOSAL.

New architectural proposals (not yet approved)

- **seating**: NEW PROPOSAL: rows are concentric arcs centred on the platform focus, so every chair faces the platform centre (no head-turn beyond the offset between focus and pulpit).
- **stage**: NEW PROPOSAL: thrust platform on the long north wall - rear band 30 m x 6.5 m against the stage wall for band / musicians plus a semicircular thrust (radius 5.0 m) carrying the pulpit 1.8 m behind its front edge; about 234 m2 at +0.90 m.
- **foyer**: NEW PROPOSAL: 12 m deep entrance foyer along the whole south long side under the same mono-pitch roof as the hall (one roof plane, external eaves gutter on the entrance facade, no box gutters). Two wide central stairs from the rear passage doors (+L02) to the foyer (client preference for two central foyer stairs, carried from Option A PL02). Sound/media booth between the stair heads, raised over the foyer, viewing over the rear passage. The two rear corners of the hall rectangle that the fan leaves free become side foyers with the toilets.
- **structure**: NEW PROPOSAL: 11 mono-pitch steel trusses on grid lines A-L, spanning grid 2 -> 4 (computed, ~50 m); foyer on separate rafters. Roof rises 5 deg towards the stage. Parallel-chord Pratt trusses, 3.0 m deep. Side ambulatories and back-of-house under low lean-to / flat roofs.
- **av**: NEW PROPOSAL: one wide LED wall above the band behind the pulpit, plus two IMAG screens on the platform-side corners angled towards the far side blocks. No delay screens are needed if every seat is within 8 x screen height (checked).
- **ambulatory**: NEW PROPOSAL (same idea as Option A): covered side ambulatories outside both hall side walls giving a step-free route at grade from the foyer to the front-corner hall doors, and covered egress.
- **rear_corners**: NEW PROPOSAL: the hall rear corners outside the fan become side foyers at grade with toilets (W female, E male) and accessible WCs, entered from the main foyer; a stair from a rear-passage door descends into each.

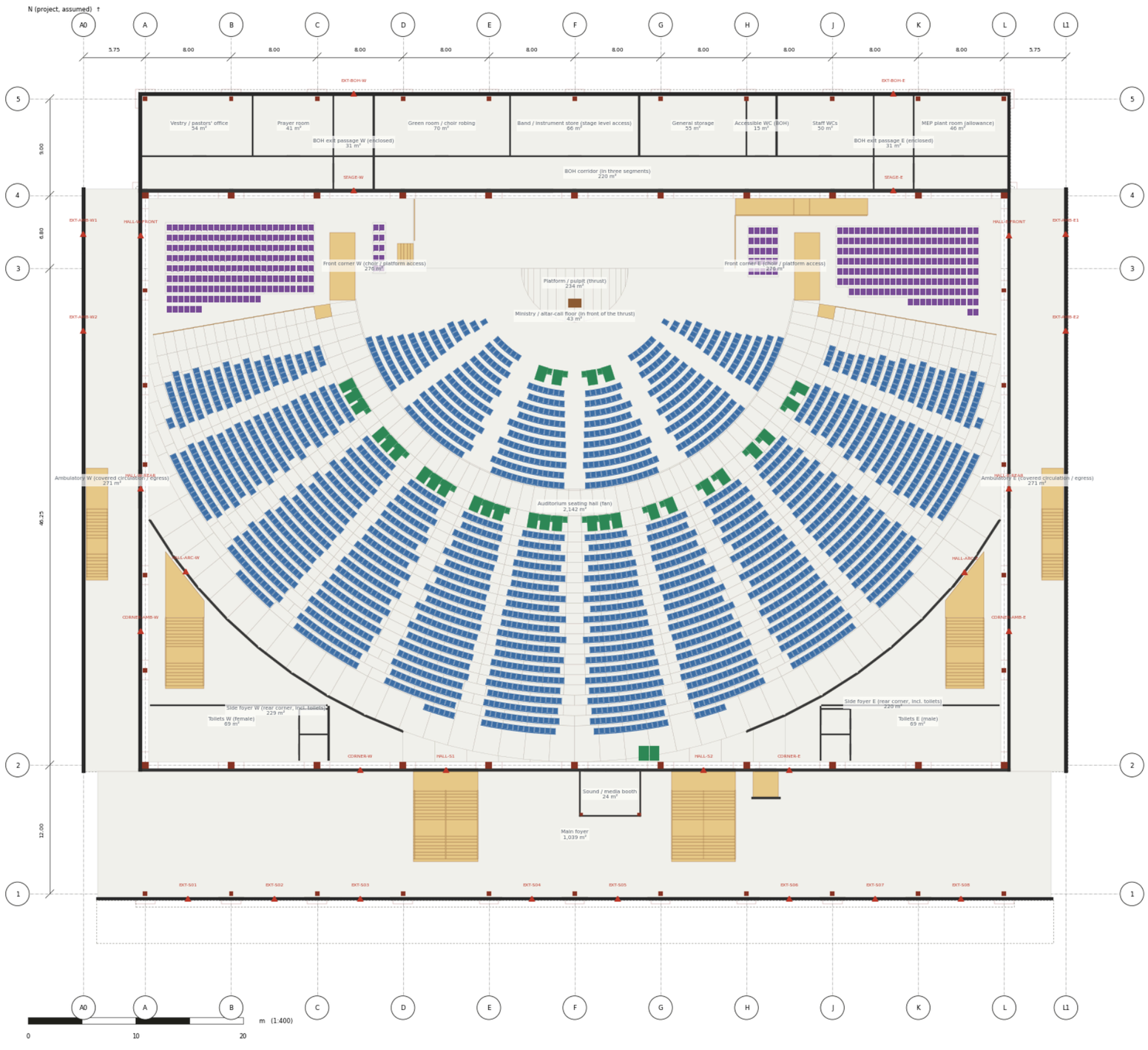
Project at a glance

Measured by comparisons/compare.py from this design's manifest (identical code for both options).

Measure	Value
Congregation seats (verified: layout, .blend, IFC, CSV)	3000
Rows	32
Eye-to-pulpit distance mean / median / max (m)	28.57 / 30.51 / 40.13
Seats within 20 m / 30 m of the pulpit	20% / 48%
Head turn to pulpit mean / max (deg)	4.0 / 18.6
Max bearing from the pulpit (deg)	80.0
Min C-value n1 / n2 (from chair objects)	0.0602 / 0.1203
Footprint (m)	91.8 x 79.55
Hall floor area / per seat (m2)	2927.9 / 0.976
Hall air volume / per seat (m3)	40190 / 13.4
Roof area (sloped, m2) / top (m)	6844.1 / 17.78
Trusses / span (m)	11 / 53.05
Columns / pad footing concrete (m3)	56 / 140.0
Hall exits / door width (m)	10 / 40.8
Wheelchair spaces / lifts	30 / 1
Choir chairs / platform area (m2)	412 / 234.1

Verification

Source	Result
Layout engine checks	{"PASS": 31, "FAIL": 0, "WARN": 2}
Re-opened .blend checks	{"PASS": 17, "FAIL": 0, "WARN": 0}
IFC validation	{"PASS": 21, "FAIL": 0, "WARN": 0}



PROJECT

3,000-Seat Church Auditorium - Option B (long-side stage) Coordination Concept

DESIGN

B-long-side-stage

MODEL REVISION

B03-44bc3ae457

DRAWING

Ground floor plan

Plan at L00, cut below the roof structure

SCALE

1:400 at A3

SHEET

B-P100

SOURCE

`blender_manifest.json sha256 356464e19ccc094f`

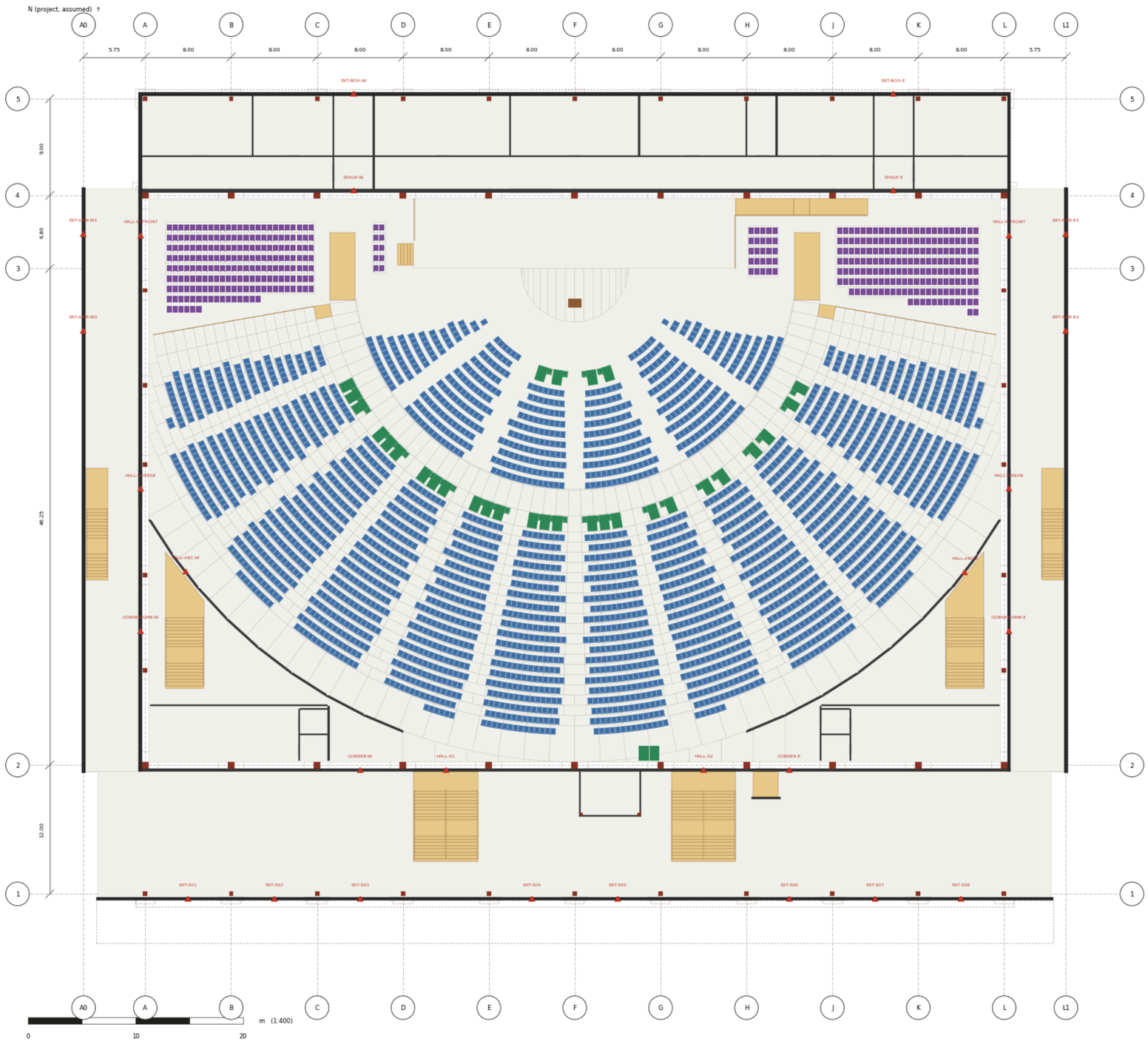
UNITS

metres; levels in m above L00

STATUS

CONCEPT - UNVERIFIED - NOT FOR CONSTRUCTION

Concept coordination drawing generated from the Blender model. Not for construction. Dimensions unverified pending professional review.



PROJECT

3,000-Seat Church Auditorium - Option B (long-side stage) Coordination Concept

DESIGN

B-long-side-stage

MODEL REVISION

B03-44bc3ae457

DRAWING

Seating, aisles and exits

Plan at L00, cut below the roof structure

SCALE

1:400 at A3

SHEET

B-P110

SOURCE

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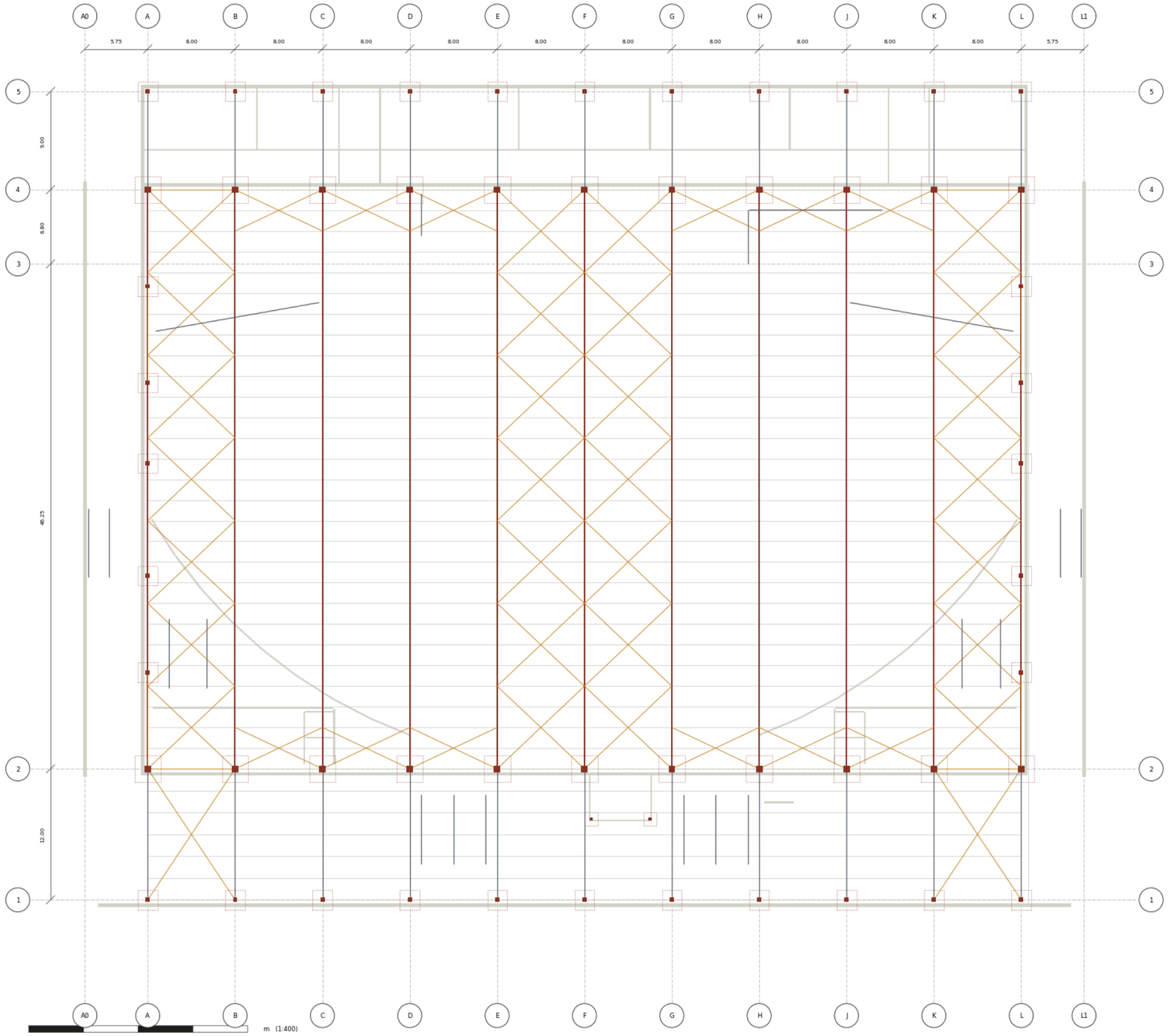
UNITS

metres; levels in m above L00

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PROJECT

3,000-Seat Church Auditorium - Option B (long-side stage) Coordination Concept

DESIGN

B-long-side-stage

MODEL REVISION

B03-44bc3ae457

DRAWING

Structural grid and roof framing

Columns, pad footings (dashed), trusses, purlins and bracing

SCALE

1:400 at A3

SHEET

B-S100

SOURCE

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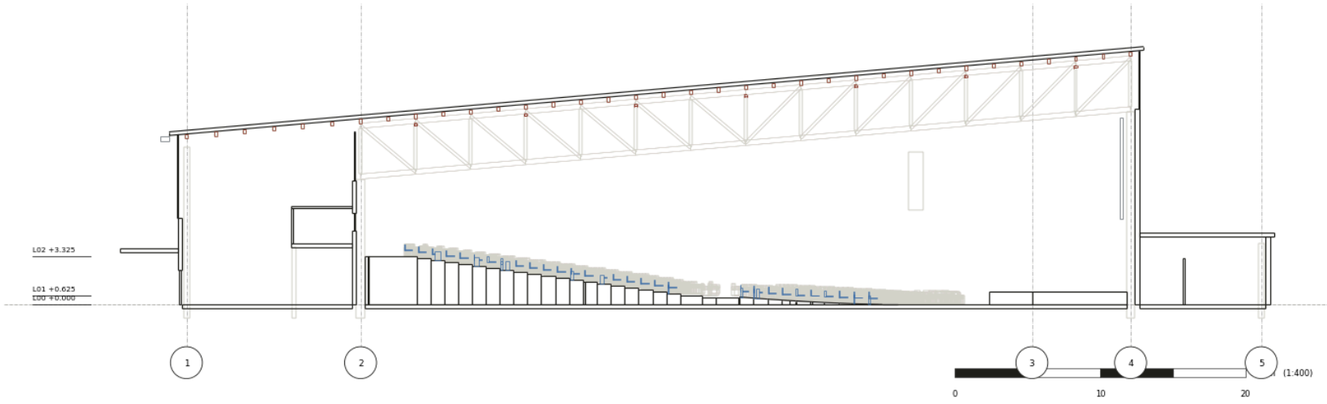
UNITS

metres; levels in m above L00

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PROJECT

3,000-Seat Church Auditorium - Option B (long-side stage) Coordination Concept

DESIGN

B-long-side-stage

MODEL REVISION

B03-44bc3ae457

DRAWING

Section A-A

Cut on x = 44.00 m, looking towards +x

SCALE

1:400 at A3

SHEET

B-S300

SOURCE

`blender_manifest.json` sha256 356464e19ccc094f

UNITS

metres; levels in m above L00

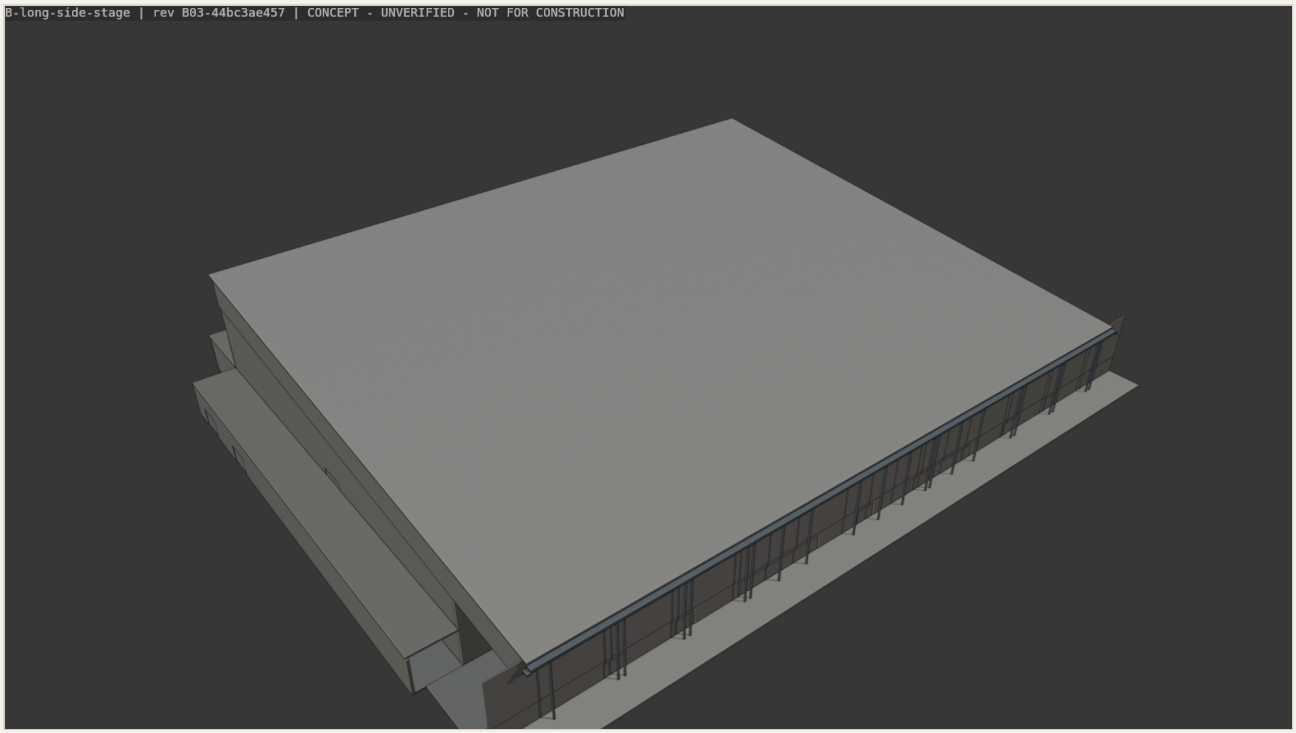
STATUS

CONCEPT - UNVERIFIED - NOT FOR CONSTRUCTION

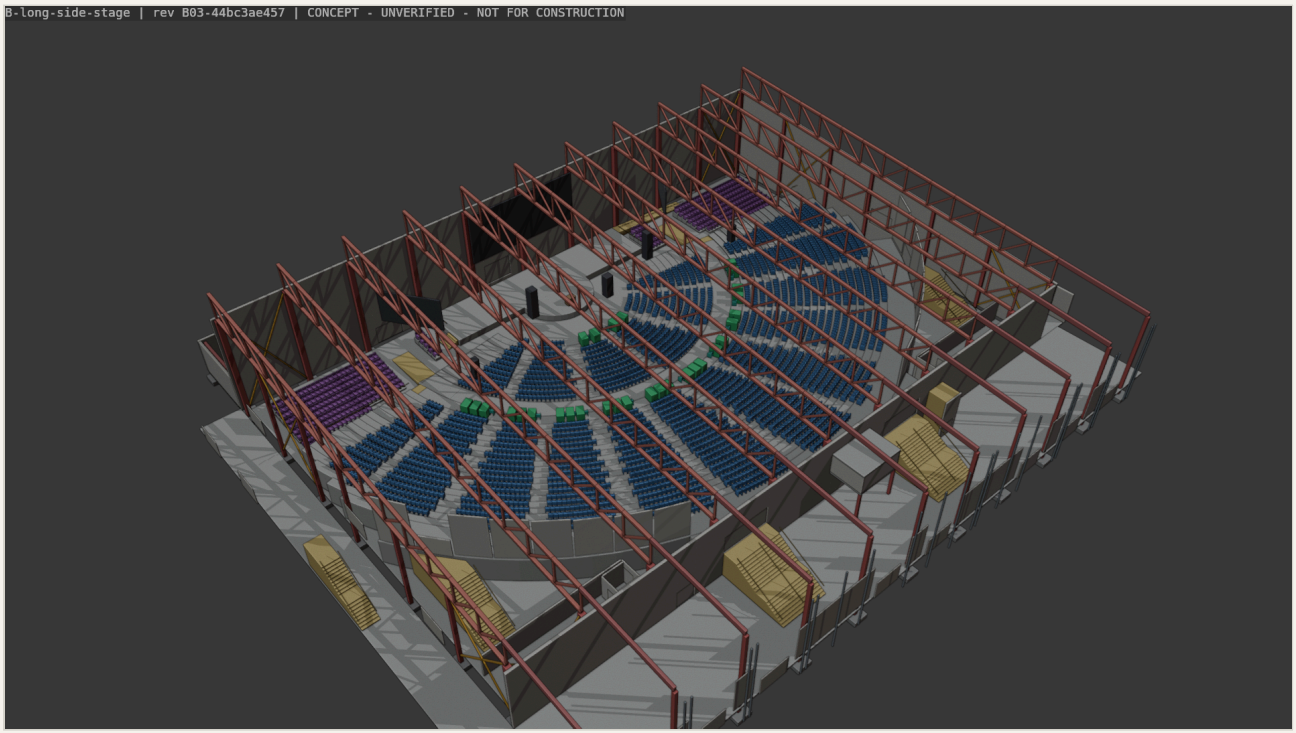
Concept coordination drawing generated from the Blender model. Not for construction. Dimensions unverified pending professional review.

Visualisations

Workbench renders of the saved model, stamped with the revision.



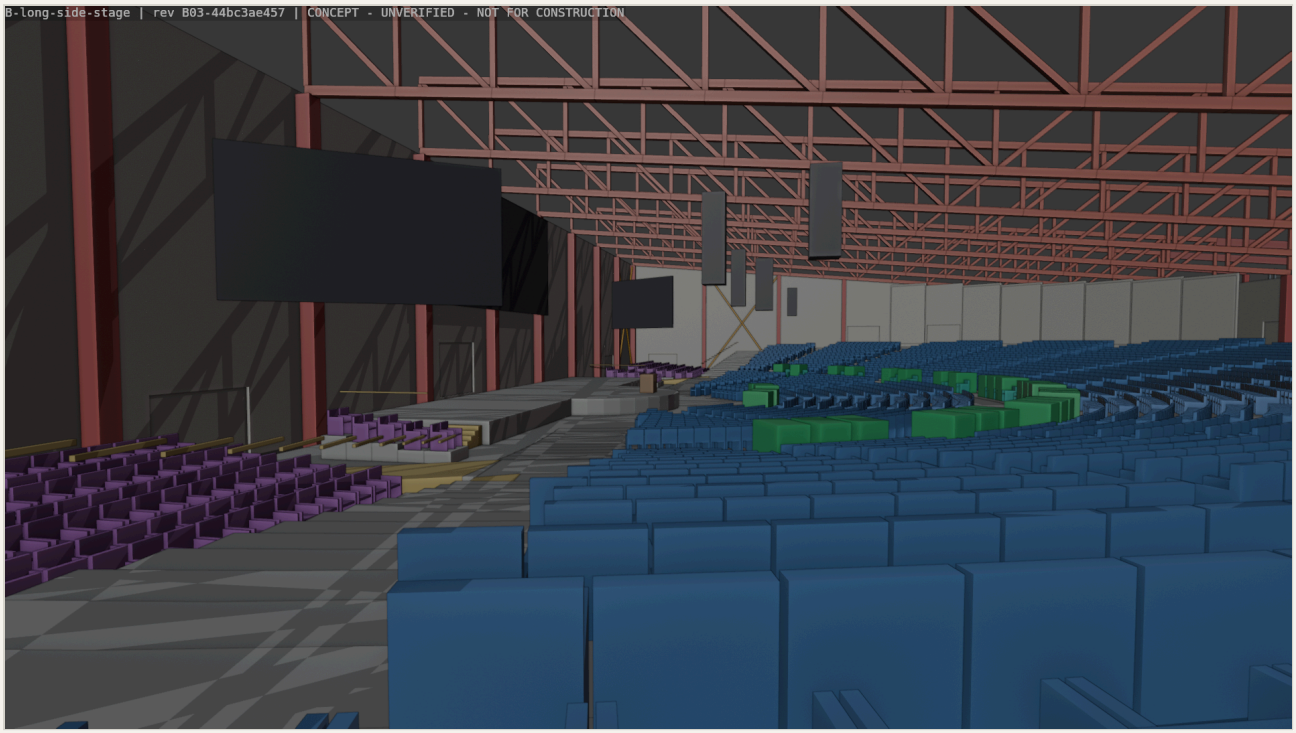
01 exterior aerial



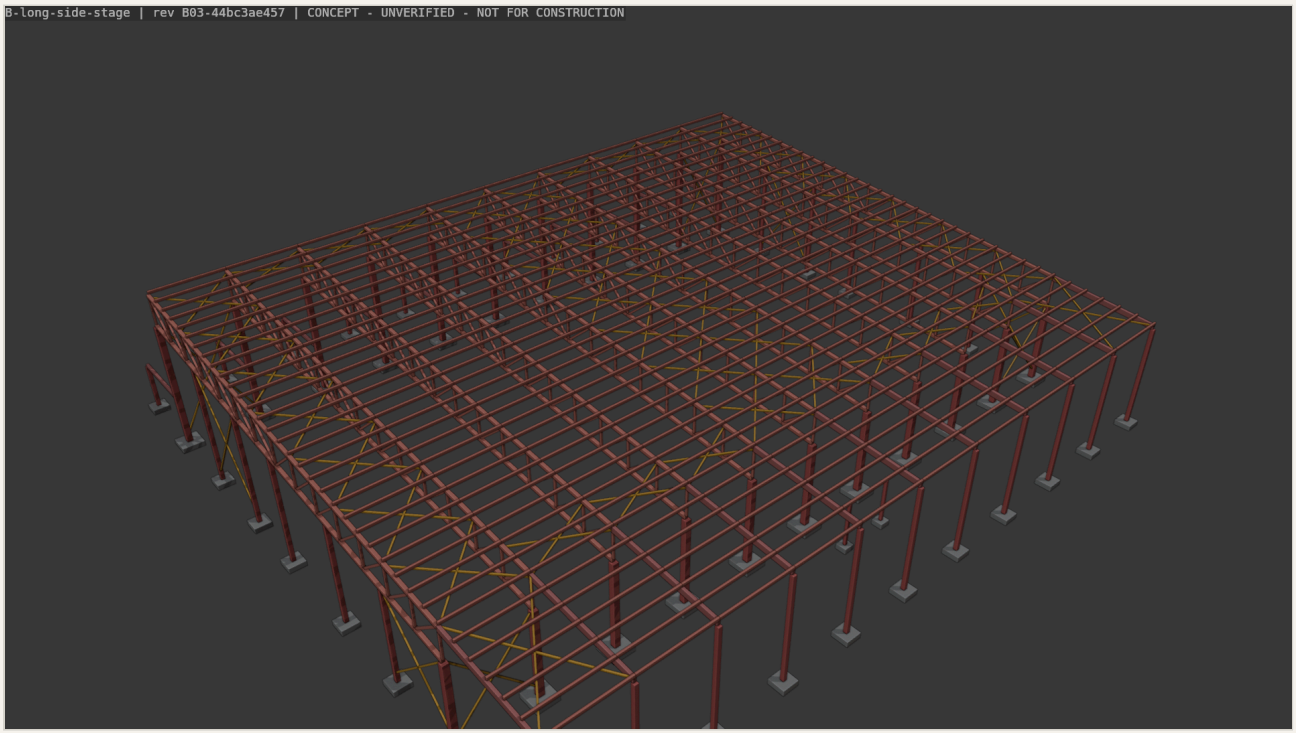
02 cutaway aerial



Visualisations (continued)



04b interior from side seats



05 structure only

Seating schedule (rows)

reports/row_schedule.csv

row	section	seats	floor	rise	radius
1	front	14	0.0	0.0	9.8
2	front	16	0.005	0.005	10.75
3	front	34	0.035	0.03	11.7
4	front	36	0.06	0.025	12.65
5	front	44	0.105	0.045	13.6
6	front	46	0.145	0.04	14.55
7	front	56	0.2	0.055	15.5
8	front	58	0.255	0.055	16.45
9	front	68	0.315	0.06	17.4
10	front	70	0.38	0.065	18.35
11	front	76	0.45	0.07	19.3
12	front	78	0.525	0.075	20.25
13	rear	88	0.76	0.235	25.1
14	rear	88	0.895	0.135	26.05
15	rear	98	1.03	0.135	27.0
16	rear	100	1.165	0.135	27.95
17	rear	110	1.3	0.135	28.9
18	rear	102	1.435	0.135	29.85
19	rear	122	1.57	0.135	30.8
20	rear	114	1.705	0.135	31.75
21	rear	134	1.84	0.135	32.7
22	rear	126	1.975	0.135	33.65
23	rear	146	2.11	0.135	34.6
24	rear	136	2.245	0.135	35.55
25	rear	148	2.38	0.135	36.5
26	rear	148	2.515	0.135	37.45
27	rear	160	2.65	0.135	38.4
28	rear	158	2.785	0.135	39.35
29	rear	144	2.92	0.135	40.3
30	rear	140	3.055	0.135	41.25
31	rear	102	3.19	0.135	42.2
32	rear	40	3.325	0.135	43.15

Egress and accessibility

reports/exit_schedule.csv, geometry_checks.json (placeholder factors, not code values)

exit_id	level	wall	clear_width	capacity_width	assigned_occupants	required_width_placeholder	final_exit
DR-HALL-W-FRONT	L00	W	3.36	3.36	129	0.645	DR-EXT-AMB-W1
DR-HALL-W-REAR	L02	W	3.36	2.0	138	1.035	DR-EXT-AMB-W2
DR-HALL-E-FRONT	L00	E	3.36	3.36	66	0.33	DR-EXT-AMB-E1
DR-HALL-E-REAR	L02	E	3.36	2.0	166	1.245	DR-EXT-AMB-E2
DR-HALL-S1	L02	S	5.6	5.6	565	4.237	DR-EXT-S03
DR-HALL-S2	L02	S	5.6	5.6	581	4.357	DR-EXT-S05
DR-STAGE-W	L00	N	3.44	3.44	597	2.985	DR-EXT-BOH-W
DR-STAGE-E	L00	N	3.44	3.44	394	1.97	DR-EXT-BOH-E
DR-HALL-ARC-W	L02	arc-W	3.36	3.36	448	3.36	DR-EXT-AMB-W2
DR-HALL-ARC-E	L02	arc-E	3.36	3.36	448	3.36	DR-EXT-AMB-E2

Egress metric (own engine)	Value
occupants_assigned_in_hall	3532
seats	3000
wheelchair	30
companions	30
choir	412
platform	60
boh_staff	30
building_total_occupants	3562
hall_exits	10
total_hall_exit_clear_width	35.52
total_required_width_placeholder	23.52
max_travel_hall	45.4
max_travel_total	58.51
mean_travel_hall	19.88
max_travel_hall_to_nearest_exit	45.4
count_nearest_travel_over_flag	459
count_travel_hall_over_flag	535
count_travel_total_over_flag	0

Risks and open issues

- **EGR-03N** (WARN): Max within-hall plan travel to the NEAREST hall exit - value {"max": 45.4, "over_flag": 459}, criterion <= 30.0 m (placeholder flag)
- **EGR-03** (WARN): Max within-hall plan travel to the queue-balanced hall exit - value {"max": 45.4, "over_flag": 535}, criterion <= 30.0 m (placeholder flag)
- Roof: 11 trusses simply supported over 53.05 m; foyer on 12.0 m rafters; member sizes, connections, splices (transport length), erection / crane plan, deflection, wind uplift and foundations are NOT designed.
- No movement joint is shown in the 80 m long building; thermal movement joints / roof expansion details TBC.
- Single eaves gutter drains the whole main roof (about 60 m of roof per metre of gutter): overflow weirs, gutter access (fall protection) and rainwater-harvesting tank capacity TBC.
- Excavation-free flat site assumed; survey, geotechnical report and storm-water strategy required (Zimbabwean rainy-season intensities).
- Egress: 535 occupants exceed the 30 m in-hall placeholder flag (max 45.4 m); the stage-side exits cross the back-of-house corridor into dedicated exit passages, which need fire separation (TBC).
- Side seats reach 80 deg bearing from the pulpit: preaching to a 160 deg fan needs a speaker who turns, and IMAG reliance (1564 seats rely on screens - placeholder 30 m unaided limit).
- Choir: 412 chairs on tiers beside the band, facing the congregation (Option A PL08: 386, facing inward). This changes the client's Option A preference (choir facing inward) - client to confirm for Option B.
- Two-thirds of the congregation leaves at the rear-passage level (+L02) by stairs; the rake height is inherent to the shared C-value targets and to a 32-row fan.
- Acoustic infill above the faceted rear-passage wall and over the rear wall between truss chords is not modelled (gap below roof).
- Booth sits 0.4-6.2 m east of the centre line because the grid-F truss passes over the axis.
- Concave faceted rear wall: absorptive / diffusive finish allowance only; focused-echo risk to be checked by an acoustician.
- Rainwater harvesting tank shown as an allowance only; capacity, treatment and overflow not designed.
- Acoustics (RT, early reflections across a wide fan), AV design, MEP, fire engineering and accessibility audits not done.

Independent review (iter-03)

comparisons/reviews/iter-03/B-*.json — QA scores, not certification.

Discipline	Score /10
accessibility	7.0
acoustics_av	6.5
aesthetics	5.0
architecture	6.3
church_functionality	7.2
constructability_maintenance	6.5
cost_drivers	5.0
life_safety	5.5
structural_engineering	7.0

Severity	Discipline	Finding
MAJOR	church_functionality	Status of iter-02 B2-ARC-03 (ministry floor): now a named space, but too small. SPC-SP02M is only 43.3 m2 and 2.5 m deep. That is about one altar-call row for a 3,000-seat congregation, about a third of A's 117 m2 SP-02F, and smaller than the about 90 m2 ring
MAJOR	church_functionality	Status of iter-02 B2-ARC-01 (choir): mostly addressed. The choir now faces the congregation on tiers rising to the stage wall (412 chairs, risers to +0.80), so it is visible and correctly oriented. However, most of it is still in the far front corners. The mai
MAJOR	architecture	Plan efficiency and entry are unresolved, and some figures got worse. The footprint has grown to 7,303 m2 (33% more than A's 5,473 m2 for the same 3,000 seats), and the truss span to 53.05 m. About two-thirds of the congregation still arrives and leaves at the
MAJOR	life_safety	This is a persistent weakness and was not addressed: two thirds of hall occupants still leave at +3.325 m and go down stairs (2,346 of 3,532, 66.4 %). Widening the side-foyer stairs only moved the bottleneck to the door. DR-HALL-ARC-W/E now carry 448 occupants
MAJOR	life_safety	Travel distances have got worse with each iteration. 535 occupants are now beyond the 30 m in-hall placeholder flag (B01 406, B02 486), and the maximum is 45.4 m (B01 41.6, B02 43.73). The direct seat-to-exit maximum is 32.8 m and the mean 16.15 m, against Opt
MAJOR	cost_drivers	Regression in cost drivers from the iter-03 changes (node-aligned purlins, deeper trusses, client-requested higher bearing). Linear roof steel is now ~6,384 m, about 2.05x Option A's 3,106 m (A's purlins are understated; see A review A-STR-01). Purlins alone a
MINOR	church_functionality	Status of iter-02 B2-ARC-05 (booth): partly addressed. The floor top is now +4.22, 0.895 m above the rear passage (+3.325), so the sightline over standing people is fixed. The booth is still 3.3 m east of the axis (x 40.4-46.2) and only 24.4 m2.
MINOR	church_functionality	Side-on seating is essentially unchanged since B02: 232 seats more than 75 deg off the pulpit (B02 228), max bearing 80 deg, and 1,564 seats rely on screens. With only 3 screens (A has 6), screen coverage for the side blocks is unverified.
MINOR	aesthetics	Status of iter-02 B2-ARC-07 (massing): somewhat improved but still weak. The taller roof (top +17.78 m) gives the interior more generous volume (13.4 m3/seat against 10.05), and the foyer glazing rose to 159.9 m2. But the exterior is still a single huge mono-p
MINOR	life_safety	The BOH escape passages are now properly enclosed, which closes iter-01 B-CB-01 and iter-02 B02-LS-03. A few details remain. The DR-BOH-X fire doors are FireExit=False with 'Single/double swing (TBC)' and an unconfirmed rating. Staff in the BOH also depend on
MINOR	accessibility	The two L02 refuge spaces sit in the open hall on the rear landing, about 1-2 m from the head of the east foyer stair, where about 580 people pass (DR-HALL-S2). They have no enclosure, are not on a protected route, and have no communication. The status text ad
MINOR	accessibility	All 60 wheelchair users and companions are still assigned only to the two stage-side exits. Those routes are now enclosed and step-free, but there is no assigned alternative if one is blocked, even though the front-corner grade exits are lightly loaded (129 an
MINOR	acoustics_av	The concave arc facets now carry modelled panels with a performance requirement, which deals with the focusing risk at concept level. Some parts of the rear wall are still bare. The door facets (-07) are untreated. The panels stop at about +8.7 to +10.1 m, bel

Revision history

Option B (long-side stage) - revision history

Revision ID = `B<nn>-<first 10 hex of SHA-256 over the declared sources>` (design.json `sources`, incl. the shared modules).
Every revision's outputs AND exact sources are archived in `archive/b-<nn>/` (rebuildable with tools/build.py).

Revision	Date	Kind	Change
B01-5f175dba4d	2026-11-02	first complete model	Wide shallow fan, 31 rows, thrust platform on the long wall, mono-pitch roof on 11 continuous trusses, ambulatories, two 6 m foyer stairs. Pre-release design iterations (seating study, clash and egress fixes) are recorded in the commit history and config `_notes`.
B02-3c1014f7b9	2026-11-02	review iteration 2	Response to iter-01 review (B 5.30/10, 1 critical blocker): BOH exit passages to north-wall final exits (blocker) + discharge-route check; stair flights <= 12 risers with landings, centre rails; 5 m ambulatories; fan +-80 deg; 4.5 m ministry floor, thrust R 4.5 m; choir on risers (377); centre-block front wheelchair places, cross-end ramps, grade-only wheelchair exits; trusses simply supported + foyer rafters, eaves wind girders, Pratt webs, larger gutter; 5 deg rear facets.
B03-44bc3ae457	2026-11-02	review iteration 3 (final)	Response to iter-02 review (B 6.08/10, 0 critical) and client comment "roof too low": truss bearing derived from the rake (>= 5.5 m clear over the rear passage, roof top +17.8 m), 3.5 m deep trusses, purlins on every node/half-panel, foyer bracing, gutter weirs; BOH exit passages enclosed from the stage wall; full-width cross-end ramps; 3.6 m side-foyer stairs (toilet blocks reshaped 17 x 5.5 m); upper-level refuges; acoustic panels on the concave wall; choir facing the congregation on tiers rising to the stage wall (412, client to confirm); platform 30 x 6.5 m + 5 m thrust (234 m2); 2.5 m ministry floor as a named space; booth +0.9 m; 16 x 7 m main screen.